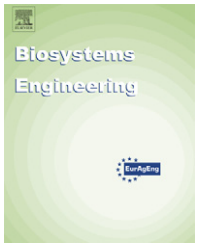
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Minimising the non-working distance travelled by machines operating in a headland field pattern

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When treating an area of field using agricultural equipment, the field is usually traversed by a series of parallel tracks using a pattern established by the experience of the operator. At the end of each track the process is constrained by the ability of the operator to distinguish the next track to be followed. The introduction of commercially available auto-steering or navigation-aid systems for agricultural machines has made it possible to upload arbitrary field pattern sequences into programmable navigational computers and for the machines to follow them with precision. This new technology also offers a new perspective for improving machine field efficiency, since not all field traversal sequences are similar in terms of total non-working distance travelled.

This paper presents an algorithmic approach towards computing traversal sequences for parallel field tracks, which improve the field efficiency of machines by minimising the total non-working distance travelled. Field coverage is expressed as the traversal of a weighted graph and the problem of finding an optimum traversing sequence is shown to be equivalent to finding the shortest route in the graph. The optimisation is formulated and solved as a binary integer programming problem. Experimental results show that by using optimum sequences, the total non-working distance can, depending on operation, be reduced by up to 50%.

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1. Introduction

Modern agricultural machines are, in theory, capable of achieving very high performance in terms of work rate and quality. A measure of machine performance during field operations is its field efficiency, E_f, which is defined (ASAE, 2005) as the ratio between the productivity of the machine under field conditions and the theoretical maximum productivity. Field efficiency is not a constant for a particular machine, but varies with the size and shape of the field, pattern of field operation, crop yield, crop moisture, and other conditions. In particular, the pattern of field operation affects very much the time lost in the field due to non-working travel (Hunt, 2001). A large portion of the non-working time occurs during turning and the strong negative influence of turning time on field efficiency has been verified experimentally in grain harvesting operations (Taylor *et al.*, 2002; Hansen *et al.*, 2003), as well as by using simulation tools (Benson *et al.*, 2002).

When turning on the headland non-working time depends on the distance travelled during turning (i.e., the length of the manoeuvre), and the mean turning speed. Some types of manoeuvre are easily executed and consequently can be executed at high speed, while other types demand skilful driving, or even reversing, resulting in lower mean speeds and requiring

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Nomenclature

- **A** set of arcs of the headland traversal graph G: $A = T \times T$
- **bl(i)** symbolises the type of the traditional where i is the number of tracks which constitute one block (see Fig. 6)
- $\mathbf{c_{ij}}$ cost that is associated to each arc A_{ij} of the graph G. This cost corresponds to the distance that a machine has to travel in order to go from the end of track i to the beginning of track j, i.e., $c_{ij} = L_{\min}(|i - j|) = L_{\min}(d_{ij})$
- $\mathbf{D_{ab}}$ Euclidean distance between the beginning of the first track of field a, and the beginning of the first track of field b
- $\mathbf{d}$ degree of a manoeuvre which is given by $d_{ij} = |i - j|$, where i is the track through which the machine exits at the beginning of the manoeuvre and j is the track the machine enters at the end of the manoeuvre
- $\mathbf{E_f}$ field efficiency
- **G** field headland traversal graph: $G = \{T, A\}$, where T is the set of graph nodes (which is also the set of tracks) and $A = T \times T$ is the set of arcs
- $\mathbf{G'}$ the extended graph G' which contains the field coverage graph G, plus the node 0 which represent the machine's initial position and the node $\|T\| + 1$ which represents the final position of the machine
- $\mathbf{J_{turn}(\sigma)}$ the total travelled distance during the turnings at the headlands for the field track coverage sequence σ (see Eq. (3))
- $\mathbf{L_{min}(d)}$ the function that gives minimum length of a manoeuvre of degree d (see Eq. (2))
- **l** the headland length
- **N** set of the nodes of the graph G': $N = T \cup \{0, \|T\| + 1\}$
- $\mathbf{p(\cdot) : T \to T}$ bijective function which for every field track, $i \in T$, returns the order in which the machine covers the ith field track
- $\mathbf{r_{min}}$ minimum turning radius of the machine
- **T** set of the field track indices. The cardinality $\|T\|$ of the set T which is equal to the total number of tracks that cover the entire field area
- **w** machine effective operating width
- $\mathbf{x_{ij}}$ decision variable of the optimisation problem which is $x_{ij} = 1$ *if and only if* immediately after visiting node i the machine travels to node j; otherwise $x_{ij} = 0$
- **Δ set** Δ_i represents the same tracks as T_i with an offset in their indexing
- ε_{ij} stochastic variable that gives the deviation of the measured manoeuvre distance from the theoretical minimum manoeuvre distance
- **Π-turn** manoeuvre type executed by an agricultural machine operating in a headland pattern (see Fig. 2b). The function $\Pi(d)$ gives the minimum lengths for a Π-turn manoeuvre of degree d (see Eq. (1))
- **T-turn** manoeuvre type executed by an agricultural machine operating in a headland pattern (see Fig. 2c)
- $\boldsymbol{\sigma}$ permutation that gives the traversal sequence of the tracks for the entire field covering. The notation σ^* represents the optimal sequence of the tracks (see Eq. (5)), while the symbol σ^{type} the sequence of the tracks according to the traditional pattern: *type*
- **Ω-turn** manoeuvre type executed by an agricultural machine operating in a headland pattern (see Fig. 2a). The function $\Omega(d)$ gives the minimum lengths for a Ω-turn manoeuvre of degree d (see Eq. (1))

larger headland areas. Furthermore, some manoeuvres unfavourably influence soil condition (Keller, 2005; Ansorge & Godwin, 2007). Vertical disturbance of the soil is caused by the weight of the machine, and horizontal disturbances are caused by changes in the direction of travel. As a consequence the headland areas constitute ''low productivity'' field areas (Witney, 1996). The controlled traffic systems can reduce the impact of the machine but only in the body of the field and not at the headlands (Bailey, 1997). Finally, the headland manoeuvres affect the fuel consumption, with longer and more complex manoeuvres requiring more fuel.

Recently, research on route planning for agricultural machines has been carried out in order to increase their field efficiency. Research has been directed towards computing the optimum decomposition of a complex-geometry field into sub-fields and the optimum driving direction in each field. Two algorithms for coverage path planning for agricultural fields were presented by Oksanen (2007); one (off-line), which uses a top–down approach to split complex-shaped fields into simple ones, and another (on-line), which uses a bottom–up approach to cover the field using *prediction* and *exhaustive search* methods. Sørensen *et al.* (2004) used a combinatorial optimisation method in order to optimise the driving pattern based on *a priori* characteristics of the field, the vehicle, and the corresponding implement. Another research direction is concerned with computing optimal tractor manoeuvres at the headlands (Vougioukas *et al.*, 2006; Oksanen, 2007). All the previous research in this area has been directed towards the use of autonomous agricultural vehicles. A survey on the topic of autonomous agricultural vehicles and related issues was given by Blackmore & Griepentrog (2006).

The goal of this work is to analyse one of the most common field patterns, the headland pattern, in order to develop an algorithmic method which minimises the non-working distance travelled by the machine. In this paper the term ''headland pattern'' refers to the complete covering, in a geometrical sense, of a field by a set of parallel tracks, or trips, which starts at one boundary of the field and terminates at the opposite boundary. Given such a headland pattern, the sequence in which a machine can traverse all the tracks during its operation is not unique. For example, the consecutive tracks covered by the machine may be adjacent (Fig. 1a) (continuous headland pattern), or non-adjacent (Fig. 1b) (alternating headland pattern).

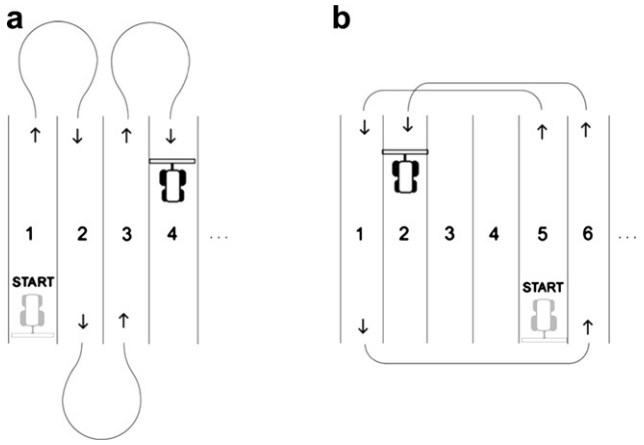


Fig. 1 – Headland pattern: (a) adjacent and (b) non-adjacent traversal; tracks are arbitrarily ordered from left to right.

In terms of total non-working distance travelled, some headland sequences are better than others. The main reason for this is that the turning distance between any two tracks depends on the turning radius of the vehicle as well as on the distance between these tracks. Additionally, the non-working distance from the initial position of the vehicle to the first track, and from the last track to its desired final position, depends on the coverage sequence.

Typically, the choice of headland traversal sequence is based on the experience of the machine operator and is strongly constrained by the presence of a human operator. A major constraint stems from the ability of the operator to distinguish the next track to be followed at the end of the track currently being taken. In some cases, this is a relatively simple task, due to the nature of the operation (e.g., harvesting or not harvesting), or due to various methods that provide traces on the field surface (soil-engaging discs, foam markers etc.). As a result, the routes followed by agricultural machines tend to form repetitions of standard patterns. This may be convenient for the operators, but it may lead to patterns which are far from optimal in terms of field efficiency, compaction, etc.

The introduction of commercially available auto-steering or navigation-aid systems for agricultural machines (Zhang *et al.*, 1999; Keicher & Seufert, 2000; Reid et al., 2000) has made it possible, in principle, to enter arbitrary field pattern sequences into programmable navigation computers and then for the machine to follow them precisely.

The objective of this paper is to present an algorithmic approach towards computing traversal sequences for headland patterns, which improve field efficiency by minimising the total non-working travelled distance. In the next section, the traversal of a headland pattern is expressed as the traversal of a weighted graph. Next, the problem of finding an optimal traversal sequence is shown to be equivalent to finding the shortest tour in the graph, also known as the TSP. The problem is formulated as a binary integer programming (BIP) problem which effectively solves the TSP problem. Finally, experimental results for some case studies are presented which demonstrate the effectiveness of the proposed approach.

2. Modelling of headland pattern traversal

The headland pattern is one of the most common field coverage patterns for agricultural machines. A given field is covered by a set of parallel tracks, or trips, which start at one boundary of the field and terminate at the opposite. It is assumed that the direction of the tracks on the field has already been determined (e.g., along its longest side) and that the field will be covered by a single machine with effective operating width w, using the headland pattern. Let $T = \{1, 2, 3, \ldots\}$ be the arbitrarily ordered set of the field track indices (e.g., from left to right). The complete covering of the field by the tracks is defined uniquely by the dimension of the field across the tracks and by the working width of the machine. The total number of tracks that cover the entire field area is given by the cardinality of the set T:

$$\frac{l}{w} \leq \|T\| \leq \lfloor \frac{l}{w} \rfloor + 1$$

where l is the headland length, w is the operating width of the machine and the symbol $\lfloor\,\rfloor$ denotes the floor function.

For any possible coverage sequence of the field, it is assumed that the entry-point of the next track and the exit-point of the current track belong to the same field headland. Then, a bijective function (one to one and onto) $p(\cdot) : T \rightarrow T$ can be defined, such that for every field track, $i \in T$, the function value $p(i)$ returns the order in which the machine covers the ith field track. For example, in the pattern that is illustrated in Fig. 1b, the values of this function are as follows: $p(5) = 1$, $p(1) = 2$, $p(6) = 3$, etc. For the pattern in Fig. 1a, the function $p(\cdot)$ is an identity function because $p(i) = i$. The inverse function $p^{-1}(\cdot) : T \rightarrow T$ gives the traversal sequence of the field tracks by the machine. For example $p^{-1}(3) = 6$, means that the 3rd track covered by the machine is the 6th field track. Hence, the traversal sequence for the entire field is given by the permutation $\sigma = \langle p^{-1}(1), p^{-1}(2), \ldots, p^{-1}(\|T\|)\rangle$. For example, the traversal sequence for the pattern in Fig. 1b is $\sigma = \langle 5, 1, 6, 2, 7, \ldots\rangle$ and for the pattern in Fig. 1a it is $\sigma = \langle 1, 2, 3, \ldots\rangle$.

2.1. *Headland manoeuvres*

Fig. 2 illustrates some of the most common manoeuvres for an agricultural machine operating in a headland pattern. These are the double round corner (Π-turn), the loop, or forward-turn (Ω-turn) and the reverse, or switch-back-turn (T-turn) (Witney, 1996; Hunt, 2001). The last two types of turn arise from the kinematic restrictions of the machine and they are executed only when Π-turns cannot be performed. Which of the two (Ω or T) will be executed is determined by factors such as the type of implement the machine caries, the operator's skill and attitude, and the available space etc. The execution of a Π-turn is restricted by the constraint $x > 2r_{\min}$, where x is the distance from the exit-point of the current track to the entry-point of the next track and $r_{\min}$ is the minimum turning radius of the machine. The farm equipment industry generally defines the minimum turning radius of a machine, as the radius of the circle within which the machine can make its shortest turn. Such a definition reports the radius of the

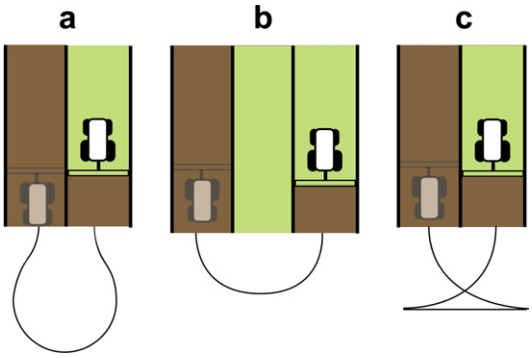


Fig. 2 – Manoeuvres in the headland turns (a) loop turn (Ω-turn), (b) double round corner (Π-turn), (c) reverse turn (T-turn).

path of an extreme part of the machine. In this work the turning radius refers to the distance from the instantaneous centre of curvature (ICC) to the midpoint between the two rear wheels, while the steer-able wheels are at their maximum steering angle (Fig. 3) (Dudek & Jenkin, 2000).

For shake of presentation simplicity only the Π-turn and the Ω-turn are considered in this work. This assumption does not affect the formulation and the solution of the optimisation problem as it will be seen later.

2.2. Non-working turning distance

We define the degree of a manoeuvre as the number $d_{ij} = |i - j|$, where i is the track through which the machine exits at the beginning of the manoeuvre and j is the track the machine enters at the manoeuvre's end. The minimum lengths for a manoeuvre of degree d, for the Ω-turn and Π-turns, respectively, can be computed by the kinematic equations of motion of a non-holonomic vehicle and are given by the following expressions (Bochtis et al., 2006):

$$\Omega(d) = r_{\min}\left(3\pi - 4\sin^{-1}\left(\frac{2r_{\min} + d\cdot w}{4r_{\min}}\right)\right)$$

and

$$\Pi(d) = d\cdot w + (\pi - 2)r_{\min} \quad (1)$$

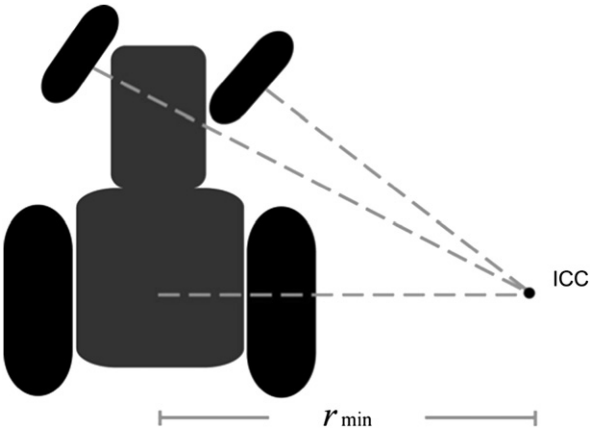


Fig. 3 – Minimum turning radius for a vehicle with Ackerman steering.

Given that a Π-turn can be executed only if $2r_{\min} \le d\cdot w$, the minimum length of a manoeuvre of degree d is given by the function:

$$L_{\min}(d) = \begin{cases} \Omega(d), & d > 2r_{\min}/w \\ \Pi(d), & d \le 2r_{\min}/w \end{cases} \quad (2)$$

note that this function gives us a theoretical minimum distance based only on vehicle dynamics under the assumptions of an "ideal" driver and non slip conditions. The measured travelled distance after executing a manoeuvre of degree d in the field would actually be $L_{\min}(d) + \varepsilon_d$. For a given combination of $r_{\min}$ and w, the term ε_d is a positive number that represents the additional distance due to factors such as the driver's skill, soil condition and tyre–soil interaction, vehicle dynamics, etc. This term cannot be modelled analytically and, therefore, it can be thought of as being a stochastic variable.

From the definitions of the manoeuvre degree and the function $p^{-1}(\cdot)$ it easily arises that the manoeuvre degree of the two tracks covered consecutively by the machine at the steps i, and $i+1$, can be written as $d_i = |p^{-1}(i+1) - p^{-1}(i)|$. Hence, for any field track coverage sequence $\sigma =< p^{-1}(1), p^{-1}(2), \ldots, p^{-1}(\|T\|) >$ the total distance travelled during the turnings at the headlands is given by:

$$J_{\text{turn}}(\sigma) = \sum_{i=1}^{\|T\|-1} L_{\min}\left(p^{-1}(i+1) - p^{-1}(i)\right) = \sum_{i=1}^{\|T\|-1} L_{\min}(d_i) \quad (3)$$

2.3. Field headland traversal graph

The problem of interest is the minimisation of the distance travelled $J_{\text{turn}}(\sigma)$ during the turning of a machine at the headland. Considering this, the length of each in field track is not important for the problem, since all tracks have to be traversed anyway. What is important is the sequence in which the tracks are processed, since the total turning distance depends on this sequence. Hence, we can represent the traversal of a field headland pattern as the traversal of an undirected weighted graph $G = \{T, A\}$ where T is the set of graph nodes and $A = T \times T$ is the set of arcs. Each node in the graph corresponds to a single track and the number of nodes is equal to the number of tracks. Traversing all field tracks is equivalent to visiting all nodes in G. Each arc, A_{ij}, $(i \neq j)$, joins node i to node j, in this sequence. Each arc A_{ij} is associated with a cost c_{ij} which corresponds to the distance that a machine has to travel in order to go from the end of track i to the beginning of track j, i.e., $c_{ij} = L_{\min}(|i - j|) = L_{\min}(d_{ij})$. Clearly, such a graph represents the coverage of a field by a machine of a certain implement width, working in a headland pattern; it does not represent the geometry, or shape of a given field. Fig. 4 illustrates simple examples of this representation. In Fig. 4a, a field is divided into four tracks based on the operating width of the implement. It is assumed that the minimum turning radius of the machine with the implement makes it possible to use a Π-turn only between tracks 1 and 4; all the other tracks can be connected only by Ω-turns. Hence the manoeuvring length is

$$L_{\min}(d) = \begin{cases} \Omega(d) & d = 1, 2 \\ \Pi(d) & d = 3 \end{cases}.$$

In Fig. 4b the field and the implement width are the same as in the previous case, but a machine with smaller minimum

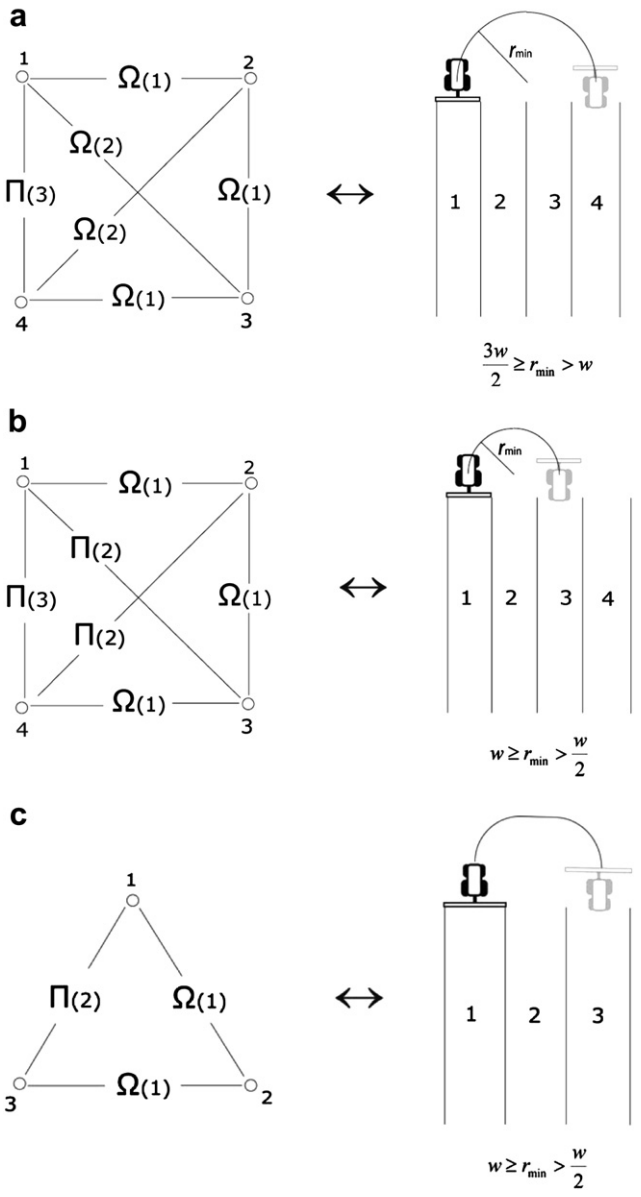


Fig. 4 – Combinations of operating width and minimum turning radius for the same field and their corresponding graphs.

turning radius is used, and so the cost of each connection between the graph vertices is changed:

$$L_{\min}(d) = \begin{cases} \Omega(d) & d = 1 \\ \Pi(d) & d = 2,3 \end{cases}.$$

Finally, if an implement with larger width is used, and a different graph is created. In Fig. 4c the graph for an implement dividing the previous field into three tracks is shown.

Using this graphical representation, the problem of covering a field in a headland pattern is equivalent to traversing all nodes of its coverage graph; different field shapes with equivalent graphs correspond to the same problem. For example, the coverage of each one of the three fields in Fig. 5, for a given machine–implement combination, is represented by the same graph and consequently corresponds to the same problem; only the turning costs may differ.

In practice, any machine that is going to operate inside a field has to have an initial location, which is its current physical position before the operation starts, and a final location which is its desired position after the operation has been completed. These locations are defined by the problem type. For example in the harvesting case these locations may correspond to the locations of a transport truck, or a silo where the harvester unloads its grain hopper. In the case of a spraying operation they could correspond to the locations where the machine refuels or simply to the barn where the sprayer parks. Therefore, the total operation of covering a set T of tracks in a headland pattern must be represented by an extended graph G' which contains the field coverage graph G, plus two additional nodes. The initial location is represented by node 0 and the final location by node $\|T\| + 1$; letting $N = T \cup \{0, \|T\| + 1\}$ be the set of the nodes of the new graph. No arc terminates at node 0, and no arc originates from node $\|T\| + 1$. The cost of connecting node 0 to any other node j is equal to the distance c_{0j} so that the machine has to move from its initial position to reach track j. Similarly, the cost of going from any node j to node $\|T\| + 1$, is equal to the driving distance c_{jN+1} from the track j to the final position.

If there is no feasible route between any pair of nodes, the cost that is associated with this arc is infinite (or a very large positive number). Finally, it is assumed that the graph contains at least one sequence of nodes connecting node 0 to node $\|T\| + 1$, with finite cost, i.e., that there is at least one feasible traversal sequence for the field.

The traversal graph can be easily extended to model the coverage of numerous separate fields, each covered by a headland pattern (see Appendix).

3. Optimisation of headland pattern traversal

For any traversal sequence $\sigma = \langle p^{-1}(1), p^{-1}(2), \ldots, p^{-1}(\|T\|)\rangle$, the total non-working distance is given by:

$$J(\sigma) = c_{0p^{-1}(1)} + J_{\text{turn}}(\sigma) + c_{p^{-1}(\|T\|)\|T\|+1}. \quad (4)$$

This distance is the sum of the total turning distance at the headlands and the distances from the initial position to the first track and from the last track to the final position. The optimal traversal sequence of the headland pattern which maximises field efficiency is the permutation σ^* which constitutes the solution of the following optimisation problem:

$$\sigma^* = \underset{\sigma}{\operatorname{argmin}} J(\sigma). \quad (5)$$

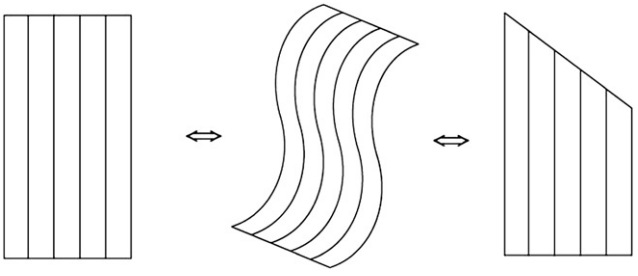


Fig. 5 – Three different fields with the same operation graph, for a given machine implement.

Solving the above problem is equivalent to finding the shortest route in the extended graph G', which starts at node 0 and finishes at node $\|T\|+1$ and visits every node only once. This is equivalent to solving the TSP, a well-known discrete or combinatorial optimisation problem (Gutin & Punnen, 2007). The TSP is an non-deterministic polynomial hard (NP-hard) problem (Garey & Johnson, 1979) and various heuristic algorithms have been developed to find near-optimal (and sometimes optimal) solutions for very large problem instances. Examples of such algorithms are simulated annealing, tabu search, genetic algorithms, etc. (Glover & Kochenberger, 2002).

Alternatively, the TSP can be formulated as a BIP problem (Wolsey, 1998). Let us define for each arc A_{ij} where $i \neq j, \quad i \neq \|T\|+1, \quad j \neq 0$ the decision variable x_{ij} which is $x_{ij}=1$ *if and only if* immediately after visiting node i the machine travels to node j; otherwise $x_{ij}=0$. Then, the minimisation of the non-working travelled distance can be expressed using the following.

For any traversal sequence $\sigma = \langle p^{-1}(1), p^{-1}(2), \ldots, p^{-1}(\|T\|)\rangle$ the total non-working distance is given by:

$$\text{Minimise}: \sum_{i\in N}\sum_{j\in N} c_{ij}x_{ij} \tag{6}$$

$$\text{Subject to}: \sum_{j\in N} x_{ij} = 1 \quad \forall i \in T \tag{7}$$

$$\sum_{j\in N} x_{0j} = 1 \tag{8}$$

$$\sum_{i\in N} x_{ih} - \sum_{j\in N} x_{hj} = 0 \quad \forall h \in T \tag{9}$$

$$\sum_{i\in N} x_{i,\|T\|+1} = 1 \tag{10}$$

$$\sum_{i\in S}\sum_{j\in S} x_{ij} \leq \|S\| - 1 \quad \forall S \subseteq N, \ \|S\| > 1 \tag{11}$$

$$x_{ij} \in \{0,1\} \quad \forall i,j \in N. \tag{12}$$

The constraints in Eq. (7) state that the track represented by a node must be traversed only once. Eq. (8) ensures that the route of the machine originates at node 0. Eq. (9) ensures that if the machine enters track h, it will exit from that track after the operation. Eq. (10) ensures that the route of the machine ends at node $\|T\|+1$. Eq. (11) is the well-known "sub-tour elimination constraint" which excludes any disjoint sub-tours from a feasible solution. Finally, Eq. (12) constrains the variables to use only binary values.

The BIP problem can be solved by various techniques, such as branch-and-bound, branch-and-cut, Lagrangean relaxation, etc. (Wolsey, 1998).

4. Materials and methods

A number of field operation experiments were performed in order to demonstrate the optimal patterns resulting from the above-mentioned methodology. The tractor that was used in all the experiments was a Ford 64.40. The trajectory of the tractor was recorded using an "AgGPS 106 Smart Antenna" EGNOS-enabled DGPS receiver from Trimble, CA, USA. The field tracks were marked and numbered at their two ends and also at intermediate points, in order to make it easier for the drivers to identify and follow them.

All the corresponding BIP problems were solved using an implementation of the Clarke–Wright savings algorithm (Clarke & Wright, 1964; Snyder & Daskin, 2004). Each optimal node sequence was transformed to a motion sequence. This is carried out using another C++ program, which in a first instance transforms the node sequence into a field work pattern and an accompanying sequence of manoeuvre types and in the second instance transforms these sequences into a sequence of Universal Transverse Mercator (UTM) path-coordinates.

In practice, the track sequence followed by an operator depends on the field, the machine, the operation and of course the experience of the operator. In order to compare the optimal sequences with a standard set of "traditional" field patterns, the established system of splitting fields into blocks was used. These blocks come about as follows: the driver starts from the first track of the block. After the completion of the operation in this track, he drives the machine into the track which he considers more "convenient" to carry out the manoeuvre, e.g., of degree a. Then, he returns to the first adjacent uncultivated track by performing a manoeuvre of degree $a-1$. Following this, he performs a manoeuvre of the same degree a as the first and so on. The direction of travel of the machine, during its operation in the initial track of a block, alternates from one block to another. Fig. 6 illustrates the patterns where blocks are formed by three tracks (a) and by five tracks (b). In the following, this type of pattern will be represented by the symbol bl(i), where i is the number of tracks which constitute one block. In order to generalise this type of patterns, we can consider that the continuous headland pattern is the degenerative case where each block is constituted by one track. So, the continuous headland pattern will be symbolised bl(1).

5. Experimental results

A number of field operation examples are presented in order to demonstrate the optimal paths that result from the above-mentioned optimisation. The implemented algorithm can easily handle problem instances with hundreds of tracks. However, for illustration purposes the examples presented here correspond to small-sized problems. The field operations took place at the farm of the Aristotle University of Thessaloniki, Northern Greece [40°32′13″N, 22°59′17″E].

5.1. *Headland traversal of a single-field*

A very simple problem consisting of covering a single-field was solved first. It corresponded to the cultivation of a small field (approximately 24 m × 30 m) using a disk harrow of working width 2.89 m. The field was divided by the harrow working width into eight tracks. The minimum turning radius of the tractor used was measured as 3.5 m. The planned operation demanded that the machine start from an initial position (a barn) on the north-west side of the field, and return to the same position after the cultivation.

A driver was asked to perform the cultivation following three different traditional headland patterns. The first was

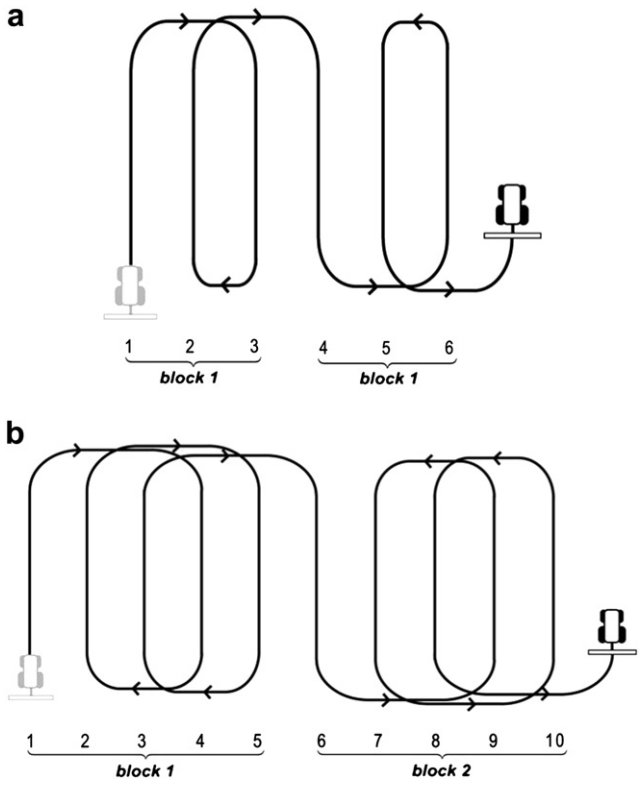


Fig. 6 – Traditional block patterns (a) bl(3) with three tracks per block and (b) bl(5) with five tracks per block.

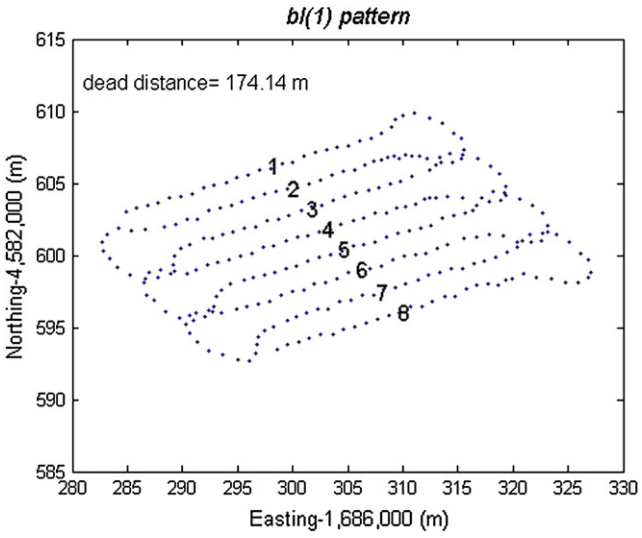


Fig. 7 – Cultivation of an eight-track field using the straight alternation pattern; the track numbers show the value of *p*(*i*).

the continuous headland pattern (Fig. 7) where $\sigma^{\mathrm{bl}(1)} = \langle 1,2,3,4,5,6,7,8\rangle$. In this operation the total dead distance was measured to be 174.14 m.

In the second pattern, the driver was asked to operate according by alternate overlapping (i.e., to avoid Ω-turns) by using the most appropriate block length, which for the particular field geometry and turning radius lead to the track sequence of the seven tracks per block: $\sigma^{\mathrm{bl}(7)} = \langle 1,5,2,6,3,7,4,8\rangle$ (Fig. 8). In this operation the total dead distance was measured as 132.35 m.

In the third pattern, the operator was asked to use five tracks per block (resulting in a sequence ⟨1,4,2,5,3,6,7,8⟩). At the end the operation the driver was forced to perform two *T*-turns in order to complete the operation (Fig. 10). In this operation the total dead distance was measured as 152.84 m.

Finally, the optimal sequence of the tracks was computed by the optimisation procedure and was found to be $\sigma^* = \langle 1,4,7,3,6,8,5,2\rangle$ (Fig. 9). The operator was asked to follow this sequence and the total dead distance for this sequence was measured as 99.65 m.

The dead distances of each of the four operations are given in Table 1. The last row shows the % savings for the dead distance achieved by adopting the optimum pattern instead of one of the driver-chosen patterns.

5.2. *Headland traversal of numerous fields*

In the next experiment the same tractor ($r_{\min} = 3.5$ m) was used to cultivate three different fields using a cultivator with operating width 2.5 m. The dimensions of the fields were approximately: field 1: 50×80 m^2, (20 tracks), field 2: 30×40 m^2, (12 tracks) and field 3: 30×70 m^2, (12 tracks). Their locations are shown in Fig. 11, together with the initial and final tractor locations, both at the depot. The three fields were represented by a single virtual field covered by a set $T = \Delta_1 \cup \Delta_2 \cup \Delta_3 = \{1,2,\ldots,44\}$ with 44 tracks.

The corresponding BIP problem was solved and the best field sequence was found to be: field 1, field 2 and field 3. The optimal track sequence for field 1, field 2 and field 3 were $\sigma_1^* = \langle 20,17,14,11,8,12,9,3,6,2,5,1,4,7,10,13,16,19,15,18\rangle$, $\sigma_2^* = \langle 1,5,11,7,3,9,2,6,10,4,8,12\rangle$ and $\sigma_3^* = \langle 2,6,10,4,8,12,9,3,7,11,5,1\rangle$, respectively. Fig. 12a–c illustrates the GPS trace from the execution of the optimal headland traversal sequence for

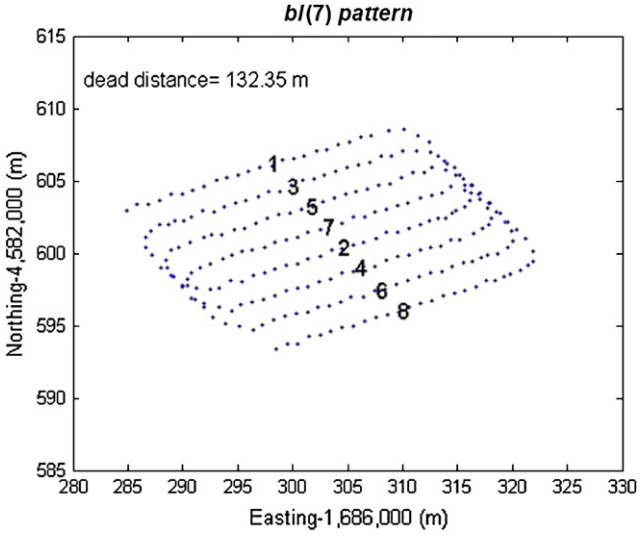


Fig. 8 – Eight-track field cultivation using the pattern bl(7); the track numbers show the value of *p*(*i*).

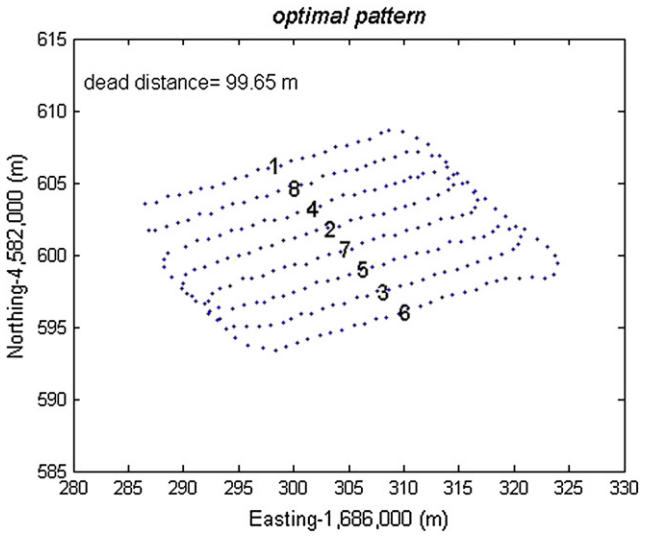


Fig. 9 – The Eight-track field cultivation using the optimal sequence; the track numbers show the value of *p*(*i*).

each field. The total path is optimal in the sense that it minimises the total non-working distance travelled during headland turnings and out-of-the field distance (from the depot to the first field, from one field to the other, and from the last field back to the depot).

When the machine operator was asked to use his own judgement, he visited the fields in the same sequence, but the track sequences he followed for covering each field were as follows: $\sigma_1^{\mathrm{dr}} = \sigma^{\mathrm{bl}(19)} = \langle 1, 11, 2, 12, 3, 13, 4, 14, 5, 15, 6, 16, 7, 17, 8, 18, 9, 19, 10, 20\rangle$ and $\sigma_2^{\mathrm{dr}} = \sigma_3^{\mathrm{dr}} = \sigma^{\mathrm{bl}(11)} = \langle 1, 7, 2, 8, 3, 9, 4, 10, 5, 11, 6, 12\rangle$. The non-working travelled distances measured during the headlands turnings for all fields and for both traversal sequences (algorithmic and operator-selected) are given in Table 2. The last column shows the % savings for the measured

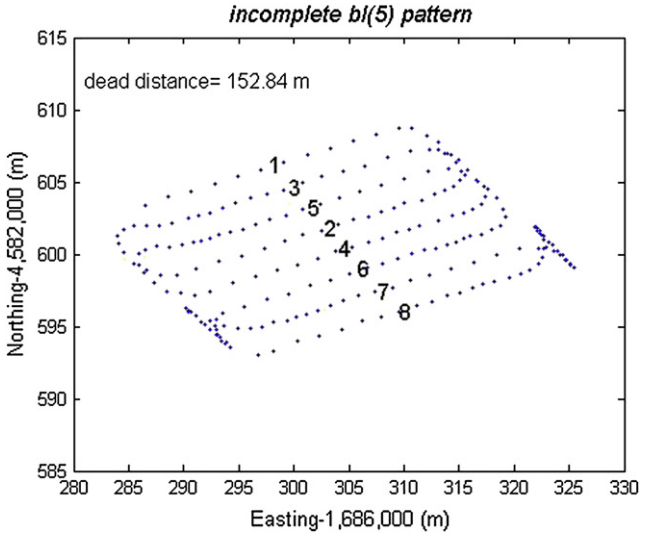


Fig. 10 – Eight-track field cultivation using the overlapping alternation pattern; the track numbers show the value of *p*(*i*).

Table 1 – Measured dead distances during turnings for four different patterns

Pattern	σ^*	$\sigma^{\mathrm{bl}(1)}$	$\sigma^{\mathrm{bl}(7)}$	$\sigma^{\mathrm{bl}(5)}$
Dead distance (m)	99.65	174.14	132.35	152.84
Savings (%)[a]	–	42.8	24.7	34.8

a $J(\sigma^{\mathrm{bl}}) - J(\sigma^*)/J(\sigma^{\mathrm{bl}}) \times 100\%$.

turning distances using the optimal and driver-chosen sequences, i.e., the ratio of the turning distance difference over the turning distance using the driver-chosen pattern.

5.3. *Comparison of theoretical and measured savings*

Given an arbitrary track traversal sequence and an optimal sequence, the turning distance savings can be computed theoretically, using Eq. (3) to compute the corresponding total turning distances. The savings reported in the previous experiments were based on measurements from a set of experiment instances. In order to compare the theoretical or predicted vs. the measured savings when using optimal patterns instead of traditional ones, three cultivations were performed on three fields A, B, C, with different operating width for each field. The operating width was 3 m in field A, 4.5 m in field B and 6 m in field C. The total width of each field was such that the tractor covered 40 tracks. Hence, field A was 120 m wide, field B 180 m, and field C 240 m. The best driver-chosen block pattern was selected for each field. Field A was covered with a pattern of 5 tracks per block: $\sigma^{\mathrm{bl}(5)} = \langle \overbrace{1, 4, 2, 5, 3}, \overbrace{6, 9, 7, 10, 8}, \ldots\rangle$, field B with three tracks per block: $\sigma^{\mathrm{bl}(3)} = \langle \overbrace{1, 3, 2}, \overbrace{4, 6, 5}, \overbrace{7, 9, 8}, \ldots\rangle$ and field C with the continuous pattern: $\sigma^{\mathrm{bl}(1)} = \langle 1, 2, 3, \ldots\rangle$. In each field the driver was asked to follow the corresponding driver-chosen block pattern and the optimal pattern computed by the optimisation.



Fig. 11 – Locations of the three dispersed fields for the cultivation operation.

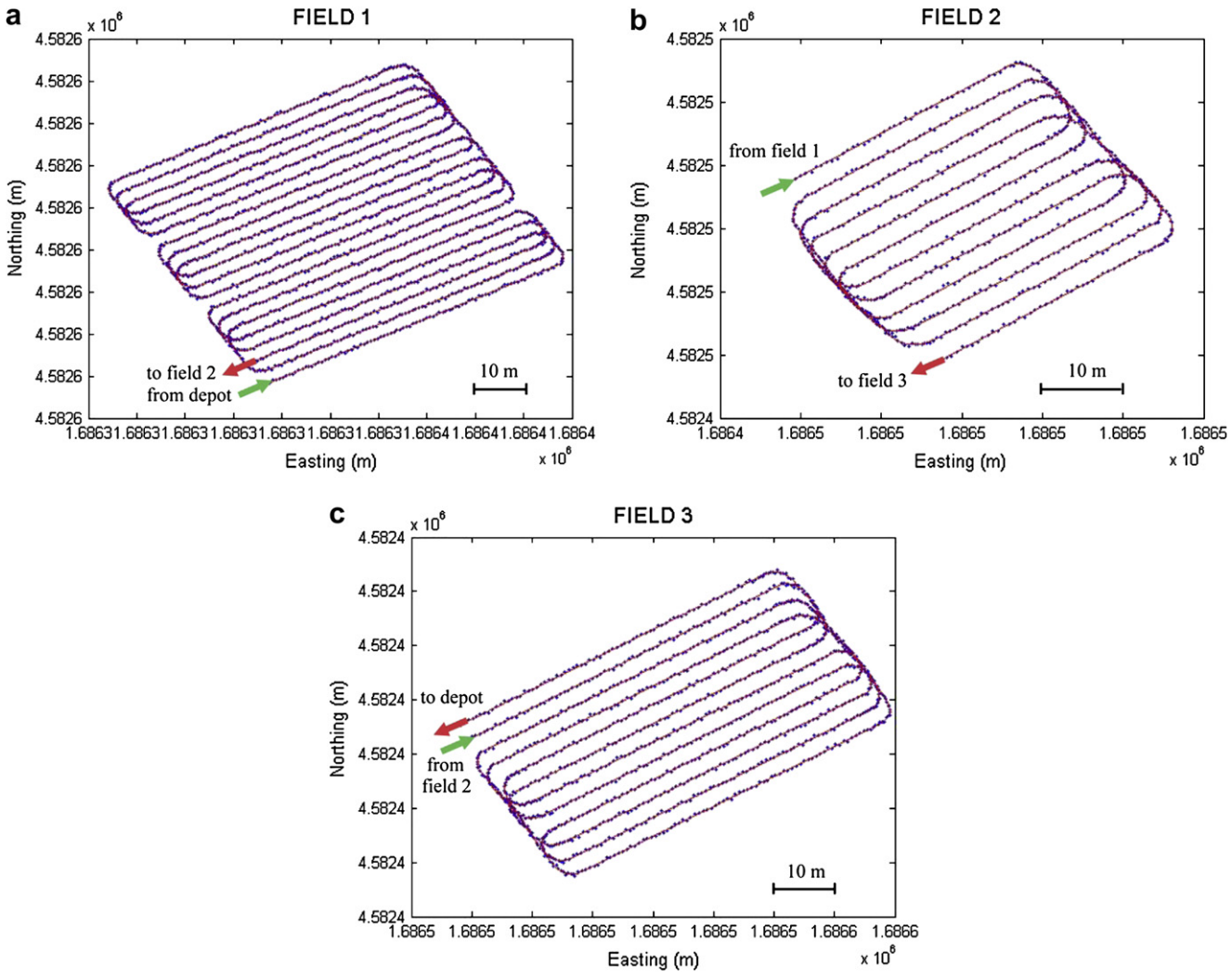


Fig. 12 – Optimal machine paths for covering (a) field 1, (b) field 2 and (c) field 3.

Table 3 presents for each experiment the dead distance measured in the field (column 4) and the corresponding theoretical distance as it is computed by Eq. (3) (column 5). Column 6 shows the % savings for the measured turning distances with the optimal and driver-chosen sequences, i.e., the ratio of the turning distance with the driver-chosen pattern. Column 7 shows the same savings percentage based on the theoretical distances (from Eq. 3).

Table 2 – Measured dead distances during turnings at the headlands for the three fields

Fields	Non-working turning distance at headlands, m		Savings, % in turning distance
	Optimal sequence, σ^*	Driver sequence, σ^{dr}	
1	258.54	565.40	54.27
2	181.57	210.20	13.62
3	176.57	242.21	27.10
Total	616.68	1017.81	39.41

6. Discussion

Experimental results showed that by using algorithmically computed optimal sequences instead of operator-selected ones, the total non-working distance can be reduced significantly. The savings on the non-working distance depend on the operation (e.g., field geometry, operating width, machine minimum turning radius) and can be as high as 50%. Furthermore, one must consider the fact that finding the most appropriate block length for the traditional overlapping pattern requires adequate experience from the operator. If the operator does not evaluate properly the combination of the headland lengths, implement width and machine kinematics, his choice will be far from the optimum. Sub-optimal patterns may necessitate corrections which will increase the total dead distance. This fact was clearly demonstrated at the third single-field operation, where the operator was asked to use five

Table 3 – Theoretical and measured dead distances and corresponding savings

Comparison	Pattern	Operating width, m	Measured dead distance, m, $J=\sum_{i=1}^{39}(L_{\min}(d_i)+\varepsilon_{d_i})$	Theoretical dead distance, m, $J=\sum_{i=1}^{39}L_{\min}(d_i)$	Savings measured, %	Savings theoretical, %
A	Optimal	3	$J(\sigma^*)=618.25$	$J(\sigma^*)=554.25$	35.84	34.66
	Traditional		$J(\sigma^{bl(5)})=963.65$	$J(\sigma^{bl(5)})=848.32$		
B	Optimal	4.5	$J(\sigma^*)=599.36$	$J(\sigma^*)=542.31$	28.95	26.70
	Traditional		$J(\sigma^{bl(3)})=843.65$	$J(\sigma^{bl(3)})=739.82$		
C	Optimal	6	$J(\sigma^*)=674.53$	$J(\sigma^*)=650.40$	32.87	29.60
	Traditional		$J(\sigma^{bl(1)})=1004.83$	$J(\sigma^{bl(1)})=923.88$		

tracks per block. This pattern proved to be inappropriate, because at the end of the field the driver was forced to perform two *T*-turns in order to complete the operation (Fig. 10). A bigger problem could arise in cases where there is not adequate space for an Ω-turn. For example, an implement such as a cultivator may reversing impractical. In such cases the driver would have to manoeuvre the machine inside the field interior.

Beyond the minimising the non-working distance when turning a machine, the algorithmically computed optimal sequences also minimise the out-of-field travelled distance. For example, when operating in the single-field experiments, the optimal sequence caused the machine to start its operation from track 1 and complete it after traversing track 2 (Fig. 9). This makes sense, since the machine had to complete the route ''barn-field-barn'' and so, it had to enter to the field and exit the field from as close as possible to the barn (track 1). This fact can also be noticed at the numerous field operations experiment (Fig. 11). As illustrated in Fig. 12a, according to the optimum path the machine had to start from the depot and enter the first field close to its southern corner. After covering field 1, it had to exit the field again close to its southern corner and drive to field 2. For the operation in field 2 the machine had to enter and exit the field from opposite sides of the field (Fig. 12b). Finally, for field 3 (Fig. 12c) the machine had to enter and exit the field from the western field tracks (tracks 1 and 2). The reason was that the starting location for the route (the last track of field 2) and the final location (the depot) were located both at the west side of the field. Notice that the entry and exit tracks for each field are computed automatically by the optimisation. In the case of the driver-chosen pattern, as it appears from the sequence σ_1^{dr}, the driver started the operation in field 1, from track 1, which was located at the north side of the field. So, the tractor had to travel across the field headland without operating. Similarly, after operating in field 3, he travelled across the headland in order to reach the depot. So, in this case a distance equal to the sum of the lengths of the two headlands (about 80 m) was travelled, more than with the optimal patterns.

A point that has to be noticed is the absence of any Ω-turns with the optimal patterns. This fact leads to two significant benefits that arise from the adoption of the optimal patterns. The first is reduced soil disturbance because smooth Π-turns result in smaller lateral forces during manoeuvring. The second benefit is a reduction in the headland area itself. Due to the larger space that is required to execute Ω-turns, the absence of this kind of turns decreases the required space at the headlands. Furthermore, the optimal solution for most problem instances is not unique, i.e., different track sequences may minimise the total turning distance. These solutions are generated randomly from the algorithms that are usually used for this kind of optimisation problems. Hence, the execution of such randomly generated optimal patterns may lead to a more ''fairly'' distributed compaction at the headlands than the repeated execution of the same pattern over the years.

Regarding the comparison of theoretical and experimental savings, as it was expected the theoretical turning distance was always shorter than the measured one, for any traversal pattern; yet, their difference was relatively small (3.6–12%). This small difference arises mainly from the driver's inability to execute perfect shaped turn and the interaction between soil and vehicle dynamics. The closeness of the theoretically and experimentally measured turning distances implies that Eq. (3) can be used to compute a reasonable approximation of the turning distances, i.e., the cumulative distance term $(\sum_{i=1}^{39}\epsilon_{d_i})$ is small. The measured savings in dead distance were very close to the theoretical ones (in the order of 1.18, 2.25 and 3.27 for the corresponding operations in fields A, B and C, respectively) because the approximation term $\sum_{i=1}^{39}\epsilon_{d_i}$ is small and it is also present in both the numerator and denominator of the savings ratio.

7. Conclusions

An algorithmic approach for computing traversal sequences for an agricultural machine operating in parallel tracks has been developed. The computed sequences are optimal in the sense that they minimise the total non-working distance travelled. Field coverage was expressed as the traversal of a weighted graph and the problem of finding an optimal traversal sequence was shown to be equivalent to finding the shortest tour in the graph. The optimisation was formulated and solved as a BIP problem.

Experimental results showed that by using algorithmically computed optimal sequences instead of operator-selected ones, the total non-working distance could be reduced by up to 50% depending on the operation. The reduction of non-working distance entails equivalent reduction of the fuel consumption as well as of the non-productive time. Moreover, apart from improved efficiency, the adoption of optimal headland patterns may also lead to reduced soil compaction in the headland area. The optimal patterns consist mainly of Π-turns and fewer – if any – Ω-turns. The smooth Π-turns

result in smaller lateral forces during manoeuvring and consequently in less compaction.

Beyond the reduction of the soil compaction at the headland area, the proposed optimal patterns reduce the headland area itself. Due to the larger space that is required for the execution of Ω-turns, the absence of this kind of turns decreases the proportion of the field area that constitutes the headland area and consequently the proportion of the low productivity area, resulting in economic benefits.

The optimal track sequences are often counter-intuitive for human operators. However, the programmable navigation-aid systems supported by the terminals or auto-steering systems which have been developed in recent years make it feasible for a vehicle to follow arbitrary optimal patterns. In addition to its application in present day navigation systems, this technique is directly applicable to route planning of autonomous agricultural vehicles.

Appendix. Extension to numerous fields

The traversal graph can be easily extended to model the coverage of numerous separate fields, each covered by a headland pattern. Consider k fields arbitrarily numbered from 1 to k, with the ith field covered by a set of tracks $T_i = \{1, 2, \ldots, \|T_i\|\}$. The union of all tracks can be thought of as a 'virtual' field which is covered by a set of tracks $T = \Delta_1 \cup \Delta_2 \cup \cdots \cup \Delta_\kappa$ where the set Δ_i represents the same tracks as T_i with an appropriate offset in their indexing, i.e.,

$$\Delta_i = \left\{ \sum_{j=1}^{i-1} \|T_j\| + 1, \sum_{j=1}^{i-1} \|T_j\| + 2, \ldots \sum_{j=1}^{i} \|T_j\| \right\}, \quad i = 1, \ldots, \kappa \tag{13}$$

for example, if $T_1 = \{1, 2, 3\}$ and $T_2 = \{1, 2\}$ then $\Delta_1 = \{1, 2, 3\}$ and $\Delta_2 = \{3 + 1, 3 + 2\} = \{4, 5\}$, respectively, and $T = \{1, 2, 3, 4, 5\}$.

The extended traversal graph G' for the virtual field is defined exactly as the graph for the single-field case. It contains all the nodes in the track set, T, the nodes which correspond to the initial and final position, and the set of arcs, A, which connect all nodes. However, the cost c_{ij} associated with arc A_{ij} must be calculated differently, depending on whether tracks i and j belong to the same field or not. If they do, then c_{ij} is equal to the turning cost $L_{\min}(d_{ij})$, just as in the single-field case. Otherwise, the cost of going from the end of track i in a field a, to the beginning of track j in a field b, depends on the relative position of the different fields a, and b, and the positions of the tracks in their respective field.

In principle, one could calculate these costs exactly based on geometry and on the dynamics of the machine. Let D_{ab} be the Euclidean distance between the beginning of the first track of field a, and the beginning of the first track of field b; obviously $D_{ab} = D_{ba}$. In typical situations, this inter-field distance is much larger than the distance between any two tracks of the same field and orders of magnitude larger than the turning radius of the machine. Therefore, for all practical purposes it should suffice to assign the costs c_{ij} based on estimates of straight-line distances. Such an estimate is given next.

When track i belongs to field a and track j belongs to field b, the cost c_{ij} is set equal to the inter-field distance D_{ab} plus two signed terms which account for the distance (in multiples of the operating width, w) of track i from the first track in field a, and the distance of track j from the first track of field b, respectively:

$$c_{ij} = D_{ab} \mp w \cdot \left[i - \left(1 + \sum_{m=1}^{a-1} \|T_m\| \right) \right] \pm w \cdot \left[j - \left(1 + \sum_{m=1}^{b-1} \|T_m\| \right) \right] \tag{14}$$

in the above expression the terms $1 + \sum_{m=1}^{x-1} \|T_m\|$ correspond to the number of the first track in Δ_x. The upper signs are used when $a < b$, i.e., field a, comes before field b in the arbitrarily numbered field sequence (e.g., $a = 2$, $b = 4$), whereas the lower signs are used when $a > b$. The sign change can be expressed algebraically as $(a - b)/|a - b|$. To summarise, the cost c_{ij} is given by:

$$c_{ij} = \begin{cases} L_{\min}(|i - j|), & \text{if } a = b \\ D_{ab} + \frac{a-b}{|a-b|} w \left\{ \left[i - \left(\sum_{m=1}^{a-1} \|T_m\| + 1 \right) \right] \right. \\ \quad \left. - \left[j - \left(\sum_{m=1}^{b-1} \|T_m\| + 1 \right) \right] \right\}, & \text{if } a \neq b. \end{cases} \tag{15}$$

Again, it is assumed that the graph contains at least one sequence of nodes connecting node 0 to node $\|T\| + 1$ with finite cost, i.e., that there is at least one feasible traversal sequence connecting all tracks in all fields.

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